

Trade-offs in Financial AI: Explainability in a Trilemma with Accuracy and Compliance

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Abstract

As Artificial Intelligence (AI) becomes increasingly embedded in financial decision-making, the opacity of complex models presents significant challenges for professionals and regulators. While the field of Explainable AI (XAI) attempts to bridge this gap, current research often reduces the implementation challenge to a binary trade-off between model accuracy and explainability. This paper argues that such a view is insufficient for the financial domain, where algorithmic choices must navigate a complex sociotechnical web of strict regulatory bounds, budget constraints, and latency requirements. Through semi-structured interviews with twenty finance professionals, ranging from C-suite executives and developers to regulators across multiple regions, this study empirically investigates how practitioners prioritize explainability relative to four competing factors: accuracy, compliance, cost, and speed. Our findings reveal that these priorities are structured not as a simple trade-off, but as a system of distinct prerequisites and constraints. Accuracy and compliance emerge as non-negotiable “hygiene factors”: without them, an AI system is viewed as a liability regardless of its transparency. Operational levers (speed and cost) serve as secondary constraints that determine practical feasibility, while ease of understanding functions as a gateway to adoption, shaping whether AI tools are trusted, used, and defensible in practice.

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JEL Codes

G21 (Banks; Depository Institutions); G28 (Government Policy and Regulation);
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Explainable Artificial Intelligence; Machine Learning; Finance; Financial Regulation

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1 Introduction

Artificial intelligence (AI) is increasingly embedded in financial decision-making, from credit scoring to green portfolio labeling. AI systems can handle large, complex inputs and produce results, but they often operate through opaque processes, hence the label “black boxes,” i.e., systems whose internal decision logic is not directly inspectable or understandable to humans. This opacity makes it challenging for professionals and oversight bodies to understand or contest algorithmic outcomes. Consequently, such obscurity hinders the adoption of reliable AI systems in the field while, conversely, risking the placement of unwarranted trust in misleading ones (Dikmen & Burns, 2022; Bernardo & Seva, 2023).

Explainable AI (XAI) is an emerging field that aims to address this problem by suggesting techniques that provide explanations for decisions made by AI (Gohel et al., 2021). A central theme in XAI research is the trade-off between explainability and accuracy (Herm et al., 2023). Essentially, the more accurate the AI model is, the harder it is to understand what constitutes the model’s decision. Yet, this trade-off is not always linear and is highly context-dependent (Goethals et al., 2022). For instance, a professional in healthcare may prioritize accuracy, while in criminal justice, explainability and accuracy may be valued equally (van der Veer et al., 2021).

In finance, this binary view is also insufficient. Considerations go beyond accuracy and explainability, since algorithmic choices might directly impact individuals’ economic opportunities (Singh & Alawat, 2023). AI products deployed in this field do not exist in a vacuum and must operate within strict bounds of latency, budget, and law. Scholars note that financial services not only face regulatory imperatives, but also operational demands regarding latency, integration of bottlenecks and model governance that favor highly performing black boxes over inherently interpretable ones (Schemmel, 2020; Sans & Zhu, 2021). For this reason, outside of computer science, AI is often conceptualized as a technology embedded within a broader “sociotechnical system,” where technical components and social structures (institutions, norms, human actors) jointly shape outcomes (Glaser et al., 2021; Kudina & Poel, 2024) and is increasingly deployed through corporate digital responsibility frameworks in management (Toth & Bluth, 2024).

Consequently, the definition of a “good” explanation becomes audience-dependent (Hadji-Misheva et al., 2021; Maxwell & Dumas, 2023). As Miller et al. (2017) argue, explanations are social processes; they are not simply the presentation of causes, but a transfer of knowledge selected to answer a specific query. Following this broader approach, we consider explainability as a desired property of AI systems that often competes not only with model performance, but also with other organizational goals and regulatory compliance requirements unique to the field of finance.

Within our broader Marie Skłodowska-Curie Actions (MSCA) DIGITAL¹ project we recognize three main groups who receive or rely on explanations in financial AI systems: 1) finance professionals; 2) regulators; and 3) the public. Although each group faces distinct explainability needs, the present paper focuses specifically on finance professionals who are the “business users” responsible for interacting with, interpreting, and defending AI-mediated decisions within financial institutions (Arrieta et al., 2020). Accordingly, we ask: *how do finance professionals prioritize explainability relative to other competing factors in practice?*

The remainder of this paper is structured as follows. Section 2 contextualizes the study within the existing body of work in explainable AI and finance. In section 3, we outline the data collection, ethical considerations, and methodological framework we use for qualitative analysis. Section 4 presents a detailed discussion of the trade-offs among the factors that shape the use of AI in finance, and section 5 concludes and suggests avenues for further research.

¹<https://www.digital-finance-msca.com/>

2 Literature Review: Layered Requirements and Competing Tensions in Financial AI

2.1 AI and the accuracy–explainability frontier

AI is an umbrella term for computational systems that perform tasks associated with human cognition, such as perception, reasoning, prediction, and language understanding. Within AI, machine learning (ML) refers to techniques that learn statistical patterns from data, while generative models, including large language models (LLMs), produce new content such as text or code in response to user prompts. In contemporary financial institutions, all three are increasingly integrated into decision-support and decision-making pipelines. The rapid expansion of data availability and computational power has accelerated the adoption of ML-driven systems, moving institutions from simple automation toward complex, adaptive algorithms capable of processing vast, semi-structured or unstructured datasets (Petropoulos et al., 2020).

These systems now permeate key domains of financial activity. In credit risk management, ML models estimate default probabilities using richer feature sets than traditional scorecards. A central development is the use of alternative data or “weak signals”—low-intensity behavioral traces that correlate with creditworthiness (e.g., device information, online behavior, or platform usage patterns) to complement bureau scores (Li et al., 2024). Berg et al.’s (2020) study of fintech lenders shows that digital footprints (such as operating system or email provider) can match or exceed AI is an umbrella term for computational systems that perform tasks associated with human cognition, such as perception, reasoning, prediction, and language understanding. Within AI, machine learning (ML) refers to techniques that learn statistical patterns from data, while generative models, including large language models (LLMs), produce new content such as text or code in response to user prompts. In contemporary financial institutions, all three are increasingly integrated into decision-support and decision-making pipelines. The rapid expansion of data availability and computational power has accelerated the adoption of ML-driven systems, moving institutions from simple automation toward complex, adaptive algorithms capable of processing vast, semi-structured or unstructured datasets (Petropoulos et al., 2020).

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In fraud detection and anti–money laundering (AML), anomaly detection and network-based models scan large transaction streams in real time to flag suspicious patterns; here speed and coverage are critical, and institutions typically adopt complex ensemble or deep models to minimize missed alerts while controlling false positives (Bussmann et al., 2020). In trading and investment, AI spans from supervised learning for signal generation to reinforcement learning and deep learning for strategy optimization. Natural Language Processing (NLP) pipelines and, increasingly, LLMs extract sentiment and events from news, social media, and financial disclosures, feeding into risk and portfolio models or powering robo-advisory and decision-support tools.

Across these areas, institutions report the same structural dilemma: the models that deliver the most predictive accuracy are typically the least transparent, intensifying demands for explainability from supervisors (e.g., central banks), boards, and internal model risk teams (Aquilina et al., 2025). From a performance perspective, additional signals and more flexible architectures shift the accuracy

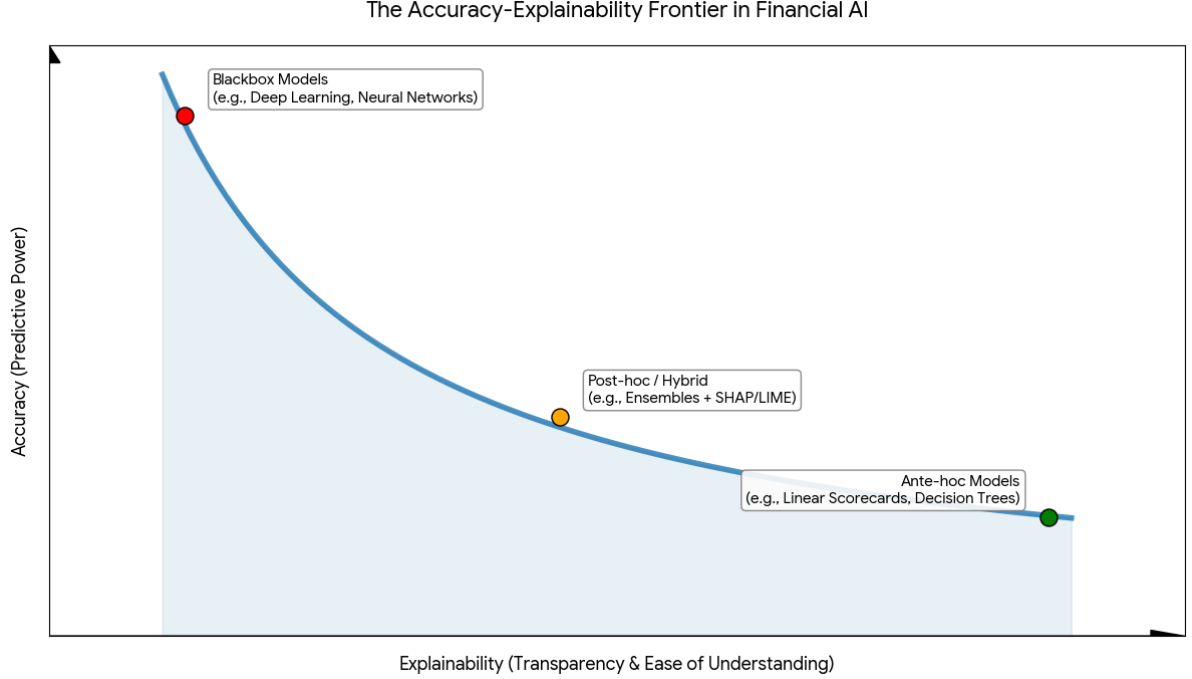


Figure 1: The Accuracy-Explainability Frontier in Financial AI

Source: Authors' construction from literature review

frontier (Figure 1) outward, improving the ability to distinguish between positive and negative classes across thresholds and enhancing calibration. At the same time, they move institutions further into “black box” territory, where decision logic is harder to articulate to customers, auditors and regulators. This tension underpins the well-known accuracy–explainability frontier in credit scoring and risk management (Bussmann et al., 2020; Hadji-Misheva et al., 2021).

To manage this opacity, financial institutions increasingly deploy Explainable AI (XAI) techniques. Comprehensive surveys distinguish between intrinsically interpretable (*ante-hoc*) models and *post-hoc* explanation methods layered on top of complex models. Ante-hoc models (e.g., linear scorecards, decision trees, rule lists) build interpretability directly into the model; they remain common in regulated credit risk because they are easy to audit but may sacrifice predictive performance on complex, high-dimensional data. Post-hoc model-agnostic explainers are now standard wherever non-linear ML models are used, notably:

- *SHAP* (*SHapley Additive exPlanations*), based on game-theoretic Shapley values, which decomposes individual predictions into feature contributions and has been widely adopted in credit risk and AML to visualize why a certain predicted score increased or decreased (Lundberg & Lee, 2017).
- *LIME* (*Local Interpretable Model-agnostic Explanations*), which approximates the complex model locally with a simpler surrogate around a prediction and has been used in financial risk prototypes to provide quick, intuitive feature-weight explanations, albeit with stability concerns (Ribeiro, Singh & Guestrin, 2016).
- *Counterfactual explanations*, which specify how a decision could change (e.g., by increasing income or reducing debt) and are directly linked to GDPR discussions about “meaningful information” and contestability.

More recently, institutions are experimenting with LLM-based explanation layers on top of existing

models. Here, LLMs, or LLMs combined with Retrieval-Augmented Generation (RAG), are used to translate SHAP values, rule outputs, or policy logic into natural-language narratives for relationship managers, call-center staff, or compliance officers. Early work on RAG for financial documents suggests that grounding LLMs in bank reports and internal documents can improve factual accuracy and user trust when answering questions about financial statements or policies (Iaroshev et al., 2024). However, these LLM explanation layers introduce a new explainability concern: the risk that explanations are plausible but not faithful to the underlying model logic (Jacovi & Goldberg, 2020).

2.2 Ease of understanding: rethinking explainability for non-technical audiences

The literature is not fully consistent in how it uses explainability and interpretability. Systematic surveys and position papers often treat interpretability as the degree to which the internal workings of a model can be understood (e.g., linear models vs. deep models), whereas explainability refers to the ability to provide reasons for specific outputs, possibly via post-hoc methods. Doshi-Velez and Kim (2017) classified evaluation strategies into application-grounded, human-grounded, and functionally grounded approaches, underscoring that explanations can be evaluated either with human users or via model-centric proxies. Later work distinguishes between computer-centered evaluation (focusing on properties like fidelity, stability, or sparsity of the explanation) and human-centered evaluation (focusing on whether people can successfully use explanations to perform their tasks) (Lopes et al., 2022).

Much of the classical XAI literature, particularly on SHAP, LIME, and counterfactuals, remains closer to computer-centric evaluation, where explanations are judged by how well they approximate the model or satisfy formal criteria such as faithfulness, rather than by how well non-technical professionals can make sense of them in context. By contrast, our empirical focus is on business users and operators of financial AI systems rather than model developers (Arrieta et al., 2020). In line with Tomsett et al.’s (2018) role-based model of interpretability, we concentrate on roles such as: operators and decision-ratifiers who must justify AI-assisted decisions to clients or internal committees; domain experts who validate model outputs against financial logic and institutional policy; risk and compliance professionals who must ensure that AI use remains within regulatory and ethical bounds.

Audience-dependent studies in financial XAI show that these stakeholders are often educated and quantitatively literate but not AI specialists. They typically do not need to understand the full internal mechanics of gradient boosting or neural networks; instead, they need explanations that are usable for decision-making, documentation, and accountability, e.g., to write a file note to a supervisor or answer a regulator’s question (Hadj-Misheva et al., 2021). For this reason, and to avoid terminological overload, we conceptualize the relevant property for our study as “ease of understanding” of the model’s outputs and reasoning, rather than formal interpretability.

2.3 A systems-view of AI and demands competing with explainability

The usual discussion of an accuracy–explainability trade-off is incomplete because it treats explainability and AI itself in isolation, rather than as one requirement within a larger context. This myopic view leads to overlooking the real impact of AI in organizations, its people, and the material environment it operates in (Glaser et al., 2021). Kuiper et al. (2019) show that finance professionals in banks and supervisory institutions evaluate AI not only at the model level, but in terms of how it fits into organizational processes, IT infrastructure, and regulatory oversight. Building on this, we distinguish four layers at which AI operates in financial institutions (Figure 2): the AI model, decision support systems, enterprise architecture, and the wider sociotechnical system. At the *AI Model Layer*, the

focus is on the component that generates predictions or decisions: the model architecture (e.g., neural networks, decision trees) and training algorithms. This is where the classic conflict between *predictive accuracy* and *model-level explainability* is most visible, and where most of the XAI literature is situated.

At the *Decision Support System (DSS) Layer*, AI outputs are embedded into tools that business users actually see, typically analytical dashboards, case-management systems, or chat-like LLM-powered interaction layers. A DSS is an interactive, computer-based information system that helps managers and professionals compile and interpret information from data, documents, and models to support semi-structured or unstructured decisions. As AI links previously siloed “information islands” into integrated decision workflows (Jia et al., 2022), the dominant concern at this layer becomes *ease of understanding and practical usefulness* of explanations. Explanations must be simple, non-technical, and context-specific so that human decision-makers can understand, trust, and act on them (Kim et al., 2020; Deo & Sontakke, 2021; Naveed, Stevens & Kern, 2022). Cognizant of this need, proposals have been developed to insert visual layers and feedback mechanisms directly into XAI models (see Bucur et al., 2025).

At the *Enterprise Architecture Layer*, the issues shift toward *speed* (or latency, i.e., the delay between input and system response) and *cost of implementation*. Enterprise architecture (EA) is an integrated, high-level description or blueprint of an organization from a business and IT perspective, used to align strategy, processes, information, applications, and technology (Saint-Louis, Morency, & Lapalme, 2019). In financial institutions, EA provides the blueprint that aligns regulatory, risk-management and customer-facing requirements with underlying data and application architectures, and is increasingly used to support digital transformation and regulatory compliance in banking (Gerber et al., 2020; Möhring et al., 2023; van de Wetering, 2021). There is a lot of incentive to invest in AI because it reduces operational staff costs and ease of decision-making as in the classic loan approval case (Christensen, 2021).

At the EA layer, budgets must cover not only model development or purchasing of existing solutions, but also whether this affects the firm’s competitiveness in the market (Kabza, 2020), and cost to compliance (Kerjriwal, 2023). In XAI, some explanation techniques, such as computing exact Shapley values, carry substantial computational costs (Mitchell, Frank & Holmes, 2022), which can directly conflict with latency requirements in time-critical domains like trading and fraud detection (Bussmann et al., 2020) or services that require near real-time responses (Hussain & Prieto, 2016). At the same time, evidence from inherently interpretable models in finance suggests that explainability does not always entail a large performance sacrifice, with some models achieving excellent risk and return metrics at a relatively low “cost of explainability” (Dessain, Bentaleb & Vinas, 2023).

Finally, at the *Sociotechnical System Layer*, AI is situated within the broader web of institutions, laws, markets, and affected stakeholders. AI through a sociotechnical perspective emphasizes that systems and social forces (e.g., values, regulations, professional norms) mutually shape each other over time (Noorman & Swierstra, 2023). This is where fairness, transparency, and systemic risk come to the forefront, because decisions made by financial AI systems can affect people and markets far beyond a single firm (Benk et al., 2022; Svetlova, 2022). These values are often encoded in national or supranational rules that financial institutions must obey. For example, the EU General Data Protection Regulation (GDPR) grants individuals affected by automated decisions the right to contest outcomes, express their views, and request human intervention (Hamon et al., 2022). Together with sectoral rules dictated in the EU AI Act and conduct regulations covering fair lending (e.g., Equal Credit Opportunity Act in the US) (Terzic, 2025) and consumer protection including investors (Lee, 2020).

Beyond individual model performance and explainability, regulators also worry about systemic vulnerabilities arising from AI. The European Parliament (2025), for instance, has highlighted the concentration risk created when many financial institutions depend on a small number of large third-party technology providers to host and develop AI models: a disruption at one provider could propagate across the system. Table 1 summarizes how the five key aspects we focus on (i.e., accuracy, ease of understanding,

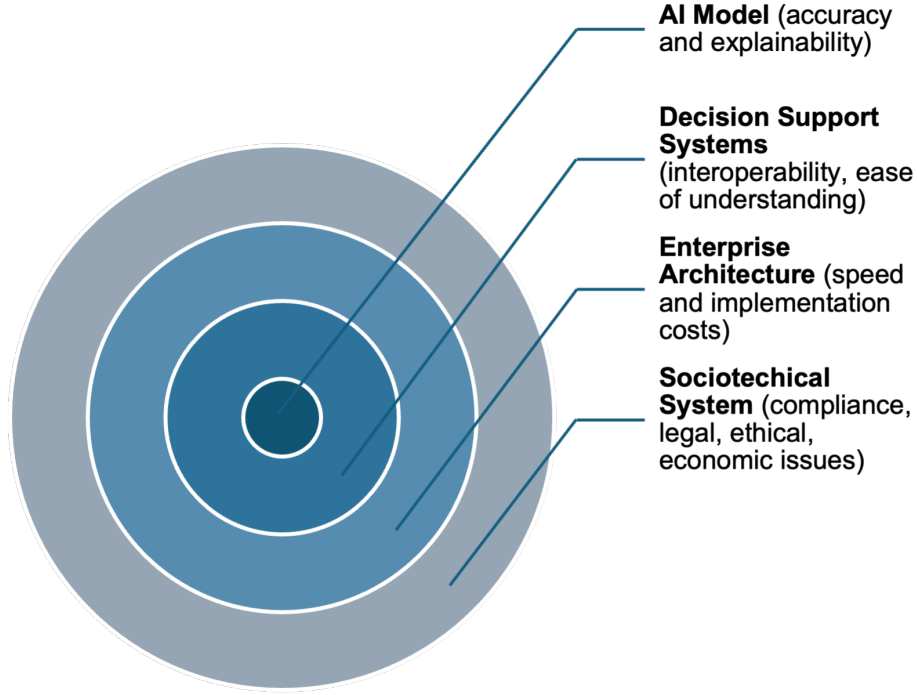


Figure 2: Considerations for AI deployment in the financial sector

Adapted from Paliwal et al. (2025)

compliance, cost, and speed) map onto these layers as competing or constraining priorities in financial AI.

Table 1: Competing priorities in financial XAI

Source: Authors' literature review.

Aspect	Definition
Accuracy	Statistical and economic performance of the model, typically assessed through predictive power against relevant benchmarks.
Ease of understanding	The degree to which decision-makers can grasp why a model produced a given output, through simple, non-technical and context-specific explanations.
Compliance	Alignment of model development and use with external regulation (e.g., GDPR, EU AI Act, fair lending rules) and internal audit and risk-management standards.
Cost	Computational and organizational resources required to develop, maintain and generate explanations, including infrastructure and staff time.
Speed or latency	Latency constraints in producing outputs and explanations, particularly in real-time or time-sensitive applications (e.g., trading, fraud detection).

These layers underscore that explainability is intertwined with multiple, sometimes competing, requirements rather than standing alone as a purely technical property. In the next section, we describe our methodology for examining how finance professionals navigate these priorities when deploying AI systems in practice.

3 Empirical Strategy

3.1 Exploratory Interviews

We conducted twenty (20) semi-structured interviews with duration ranging from 30–60 minutes between April and November 2025. Our sample size aligns with qualitative research literature indicating that

thematic saturation of codes and meanings in similar studies is often achieved within 9–17 interviews (Hennink & Kaiser, 2022). The full interview guide was designed to support multiple papers and included broader questions on XAI expectations, explanation formats, and model interaction. However, only a subset of the guide was relevant to the present trade-off analysis. Accordingly, the analysis in this paper draws on three short segments of the interview:

3.1.1 Role and Context

- Can you briefly describe your role?
- Do you currently interact with AI or machine learning systems in your work? In what way?
- Are you mainly using outputs, developing/validating models, or overseeing projects that involve AI?

3.1.2 Relevance of Explanation to Work Tasks

- For the AI systems you work with, are explanations or justifications available in any form? What does that look like in practice?
- Are there tasks where explanations matter more to you?

3.1.3 Trade-off Ranking Task

Participants were presented with five competing considerations for AI deployment and asked to rank them from most to least important:

1. Accuracy
 2. Ease of understanding
 3. Compliance
 4. Cost
 5. Speed
- Could you walk me through why you put them in this order?
 - What made you place [factor X] above [factor Y]?

Participants were presented with five competing considerations for AI deployment (accuracy, ease of understanding, compliance, cost, and speed) and asked to rank them from most to least important. The items and their definitions were derived from the literature review (Table 1). Building on Habiba et al. (2024), who discuss the accuracy–explainability trade-off for ML practitioners, we extend the focus to a broader set of operational factors and investigate how finance professionals themselves prioritize and justify these trade-offs. After ranking, participants were invited to explain their ordering and to describe concrete situations in which they would change or defend their priorities.

Although this design generates ordinal ranking data, we do not perform any statistical on the rankings. This is for two reasons. First, the study is explicitly exploratory and qualitative: the rankings are used as elicitation devices to structure reflection and to prompt rich accounts of how trade-offs are experienced in practice, not as endpoints for inferential testing. Second, with a small, purposively selected sample and purely ordinal responses, conventional statistical techniques would risk over-interpreting highly contextual preferences and suggesting a level of precision that the data does not warrant. Accordingly, we

treat the rankings analytically as prompts for the participants’ narrative justifications. The primary evidence for our claims lies in these justifications and examples, rather than in numerical properties of the rank orders themselves.

3.2 Participant Recruitment

The participants (Figure 3) The participants were selected through purposive and snowball sampling, beginning with initial contacts within the MSCA DIGITAL network and connections on various social media platforms. The recruitment process targeted professionals with demonstrable experience in the application or oversight of AI systems in financial contexts. Specifically, the inclusion criteria required that each participant:

- is currently or has previously been employed in the financial sector, or in an auxiliary business primarily serving financial institutions (e.g., consulting or auditing firms, or technology vendors); and
- has directly interacted with, operated, or contributed to the development of an AI-based system or application for a period of at least six (6) consecutive months, either in a full-time or project-based capacity, i.e., as external consultants who work intensively, but may do or have done so for a limited period for a financial institution.

The first criterion focuses on the practical, real-world employment context within the sector, not academic background, to ensure participants have firsthand experience with the application of AI in the financial environment. We still included developers in the second criteria because their role and workflow usually cover systematic logging of issues in software deployment that are reported by their users. These issues are indicative of the explainability needs of our target audience. The snowball approach allowed for a targeted but diverse pool of respondents from banks, AI vendors, electronic money issuers (EMI) and the consultancies providing auxiliary services to financial institutions. The participants’ backgrounds are summarized in Figure 3, referred to anonymously as P1-P20 in the results section (see Appendix A for the full list).

Because the study is an EU-funded project, the sample is skewed toward European institutions. This introduces a geographic bias, as EU participants operate within stricter regulatory environments (e.g., GDPR, EU AI Act), which may shape their views on explainability more strongly than in other regions. Meanwhile, the gender distribution (75% men, 25% women) reflects broader underrepresentation of women in AI and financial technology, particularly in technical and leadership roles, and should be interpreted as a structural industry pattern rather than a sampling choice.

3.3 Data Analysis

Qualitative analysis of interview transcripts was performed independently by two researchers employing complementary coding approaches to enhance analytical rigor. One researcher used holistic coding, combining a deductive lens (guided by the trade-offs in Table 1) with an inductive lens that examined full segments of text to capture broad themes and narratives. The other researcher drew on the Gioia method as an organizing framework, distinguishing between more descriptive, participant-centered codes and higher-level, researcher-generated themes, without constructing a formal first-order/second-order data structure. After independently completing their coding rounds, the researchers collaboratively compared and merged their codebooks, arriving at a shared set of thematic categories used to structure the presentation of the results. While the coding was informed by Gioia-style distinctions between first-order and second-order concepts, the findings are presented in terms of these consolidated thematic

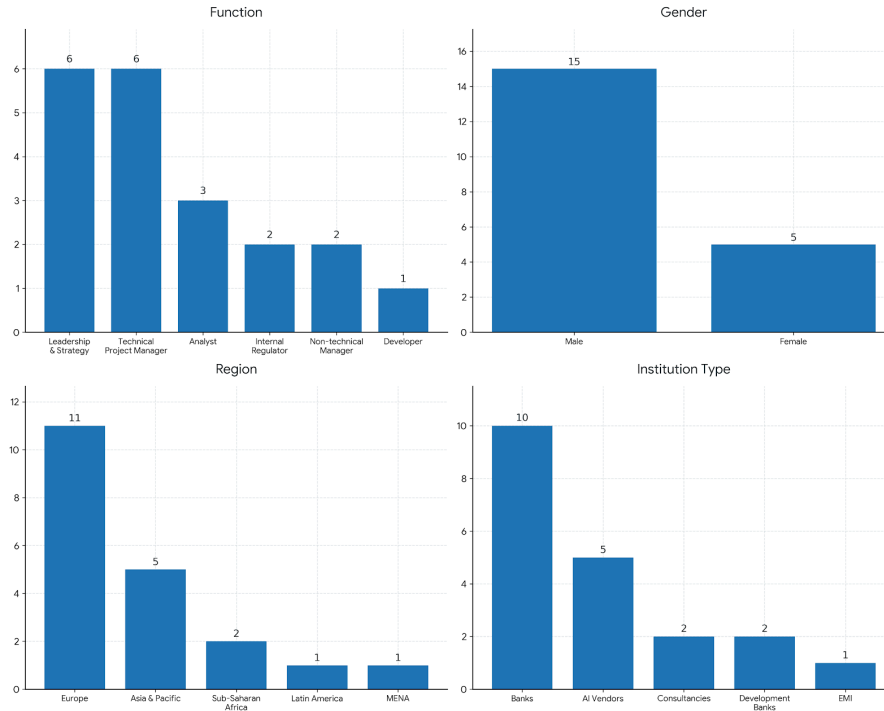


Figure 3: Summary of Interview Participants Demographics

Source: Authors' compilation

categories rather than a full Gioia data structure. This dual-coding and merging process maximizes coverage and depth in capturing the complexity of finance professionals' views.

3.4 Ethics Statement

To ensure ethical conduct and participant privacy, all interviews were conducted with informed consent. The study's purpose and participants' rights were outlined in the information sheet, including their right to withdraw at any time. The study has also been reviewed and approved by the Ethics Committee for Research Involving Human Subjects in Biomedical Fields of the University of Naples Federico II, on September 25, 2025, with reference number PG/2025/0133225.

4 AI Deployment Trade-offs as Perceived by Finance Professionals

The interviews illustrate how finance professionals navigate this XAI trilemma in practice: they treat accuracy and compliance as prerequisites, while adjusting cost, speed, and ease of understanding within those hard bounds.

4.1 Accuracy as Prerequisite to Deployment

The rationale for accuracy being a top priority is directly tied to the functional purpose and credibility of the AI system. The respondents' perspective is grounded in the understanding that an AI system is worthless if it is not correct. When an AI fails to provide correct, factually grounded information, or when it presents incorrect information with unearned confidence, it fundamentally erodes not only its utility, but also the user's trust. As P5 puts it, *"if the answer is not accurate... he or she [the client]*

would never use the tool again... trust is very much crucial. And it's the credibility we have as a research house."

Participants conceptualized accuracy differently depending on whether they were working with classical machine learning models or generative AI systems. In classical ML contexts (e.g., credit scoring models or AML transaction monitoring), accuracy was linked to performance metrics, i.e., reducing false positives and false negatives, stabilizing model behavior over time, and ensuring reliability across monitoring cycles. P8 described quarterly monitoring using R-squared (coefficient of determination) and absolute error metrics to identify issues such as a *"dip in the input data"* that could undermine outputs.

By contrast, in generative AI systems (e.g., internal Q&A assistants with LLM-assisted responses or RAG-based document search tools), accuracy was defined as factual faithfulness to source documents rather than statistical performance. P2 described accuracy as ensuring that the answer *"really, really comes from the source, not that it was somehow generated"*, supported by automated open-source frameworks that check whether outputs match original materials. Developers also cautioned that LLMs can be *"overconfident with the answer,"* resulting in a contradictory effect: the human-readable language of the interface helps users explain the model (a feature P17 appreciated), yet, as P16 warned, the explanation may ultimately bear little relation to the actual input, an issue that has been found pervasive in LLMs (Jacovi & Goldberg, 2020). Accuracy was prioritized whenever errors risked undermining explainability, confusing users, or exposing the institution to regulatory or reputational consequences (see Table 2).

Table 2: Cases for Prioritization of Accuracy

Source: Authors' transcripts		
Factor	Interplay Analysis	Supporting Quotes
Accuracy vs. Ease of Understanding	Accuracy of the explanation is necessary to validate the underlying model's performance; inaccuracy leads to misleading outputs.	"if the explanation isn't accurate, then it's just a double issue... it creates some kind of strange mix of false positives and negatives... I don't know what even was happening" (P2).
Accuracy vs. Compliance	Compliance is endangered if the system is inaccurate, leading to severe institutional consequences.	"If the AI produces 'errors or false data,' the financial institution is 'liable' to the regulator, resulting in penalty and reputational damage" (P14). Factualness is essential to "validate whether the internal guidelines is [are] meeting external guidance requirements (sic)" (P2).

Meanwhile, accuracy is seen as a secondary concern because models that are allowed have *"super stable and predictable"* performance and as such, are considered a "solved problem (P8).

4.2 Compliance as a Non-Negotiable "Hygiene Factor"

Using highly predictive, complex models leads to a contradiction with the regulatory demand for guaranteed interpretability. This reality was described viscerally by one consultant (P11) who certifies AI systems, noting that financial services are *"regulated to death,"* making compliance an absolute, *"unbreachable barrier to entry"* where *"you would just not get the approval if there [were] no explainability."* The ability to audit an AI's process and explain its decisions to regulatory bodies is thus not a secondary feature but a core design requirement. Dubbed by a participant (P17) as *"big compliance machines,"* banks face multiple regulatory mandates, especially in the Eurozone. For example, as reported by P7 and

P8, the European Central Bank (ECB) and European Banking Authority (EBA) do not allow black box models (such as neural networks or complex boosting techniques) for use in decision-making processes because they imply “*unpredictability and the risk involved.*”

An exception was noted in an application in Asia in line with Li et al. (2024)’s application of alternative credit scoring model. P6 described a major project where their team used non-traditional variables to evaluate clients for pre-approvals. P6’s model was highly successful, enabling the bank to grow its client base significantly while operating within its acceptable risk appetite. This supports the finding by Cornelli et al. (2020) that fintech and big-tech credit volumes are larger in countries where banking regulation is less stringent, indicating that lighter regimes facilitate faster rollout of alternative, often opaque, credit technologies. As described by P9 and P14 regarding the region’s slow uptake, the use of automated decisions remains a “gray area” (P14), i.e., neither actively promoted nor explicitly prohibited. P9 clarifies that if the market were more mature, regulators would have been asking “*how does the model work?*” but since the “*regime here is still weak. . . I don’t really get the questions.*”

In Table 3, we present cases when compliance is preferred over accuracy and ease of understanding.

Table 3: Cases for Prioritization of Compliance

Source: Authors’ transcripts		
Factor	Relationship	Supporting Quotes
Compliance vs Others	Compliance is a non-negotiable prerequisite.	“I just put alignment with regulations first [...] So it is a given anyhow.” (P2) “Without having compliant processes [...] So it’s quite on top.” (P5)
Compliance vs. Ease of Understanding	Compliance requires satisfying auditors and regulators [...]	“The explainability [...] for the regulator mostly.” (P10)

Notably, even for those who rank compliance low, this does not imply that compliance is unimportant. Rather, they consider it a “given” (P2) at the company level, making it a “hygiene factor,” a basic precondition that must be satisfied before other factors (like accuracy or speed) are even considered (P2, P11, P6, P17). By using a “*compliance by design*” (P13) approach, so that “*whatever regulations that come. . . you can’t run far (from complying) because your system is already doing the right thing.*” P1, on the other hand, ranked compliance last because the AI prototype tool was already limiting the scope of information provided internally.

4.3 Ease of Understanding as Gateway to Adoption

A recurring theme was the significant challenge of translating complex technical XAI metrics into meaningful, role-specific insights, aligning with previous suggestions from Schmude (2021), Hadji-Misheva et al. (2021), Maxwell and Dumas (2023), among others. While data scientists have tools to examine model internals, their outputs are often incomprehensible to the business stakeholders who must act on the model’s recommendations.

A notable case is that of P11, who structures AI autonomy into four levels and the associated explanation of rigor scales with the level of risk and the depth of the audit trail required. Their goal is to make the audit trail “*as humanly readable as possible,*” but this is complicated by the technical nature of the underlying processes. In practice, the level of technical detail of the explanations depends on who the explanation is for and how far back in the decision process one needs to look. For an end

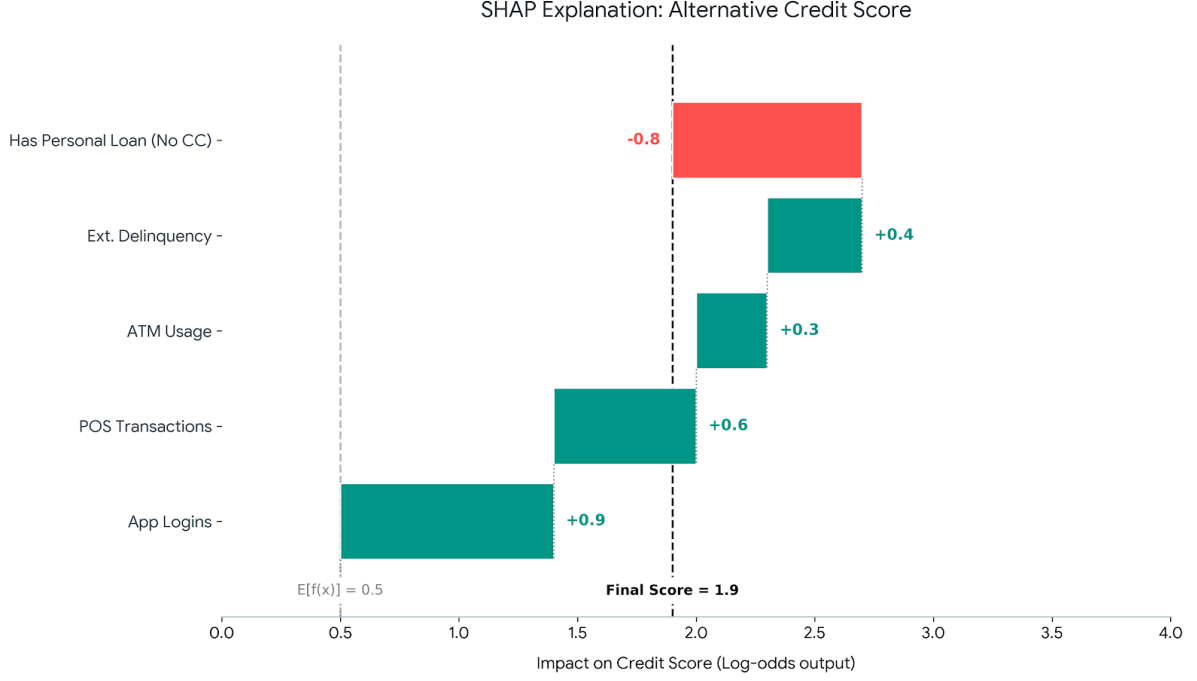


Figure 4: Reconstruction of P6’s Alternative Credit Scoring Model

Source: Authors. Note: While the specific log-odds values are for illustrative purposes only, the feature set and directionality are empirically grounded in the interview data. Specifically, it visualizes the inclusion of non-traditional behavioral traces such as App Logins and (Automated Teller Machine) ATM Usage, alongside the counter-intuitive finding that “Has Personal Loan (No CC)” acted as a negative predictor.

user at the final step, the explanation is intentionally simple (for example, “*no, because your balance is too low*”). When one traces the decision backwards through earlier steps in the pipeline, the explanation must become more technical, because the target audience shifts to developers or auditors who need to understand the underlying mechanics.

Multiple professionals noted that standard XAI tools like SHAP, which generates feature importance values, are often not “*intuitive enough towards a user*” (P6, P10), contrary to the findings of Sabharwal et al. (2025) where it increased trust of finance managers. P6 recalled the experience of presenting SHAP plots to a business team: “*The business would not understand it. They were so used to the regression coefficients*” (P6). Figure 4 provides a reconstruction of this scenario based on P6’s interview. It displays the specific mix of behavioral traces (e.g., Point-of-Sale transactions, Automated Teller Machine or ATM usage) and financial history that the data science team had to defend. P6 had to devise a “*translation layer*,” manually converting the visual data into narrative categories (linear, inverse, or complex) that matched the business unit’s mental models. Conversely, when the intended audience for the explanation possesses a high level of technical or domain expertise, ease of understanding is secondary.

This creates an *organizational* need to convert complex technical metrics into meaningful, user-centric narratives that are tailored to the audience (P6, P11). Professionals are actively taking steps and seeking methods to bridge this gap. For example, they use other AI tools like LLMs “*to make these Shapley values just more humanly understandable*” (P10). To know whether it works and to ensure accuracy, P16 emphasizes the need for constant manual checking. The interviews revealed several cases where ease of understanding is put ahead of other desiderata such as accuracy, speed, or even formal compliance. Table 4 illustrates two such situations.

Table 4: Cases for Prioritization of Ease of Understanding

Source: Authors’ transcripts		
Factor	Relationship	Supporting Quotes
Ease vs. Others	If an explanation cannot be understood, its accuracy is irrelevant.	“if I cannot communicate what I’m doing [...] no one will consume it.” (P6)
Ease vs. Accuracy	Participants described a willingness to accept some loss [...]	“I have an extremely performant model [...] perfectly explainable [...]” (P10)

4.4 Operational Levers as Secondary Factors

Though ranked lower in importance by many participants, factors such as cost and speed act as crucial constraints that shape the practical feasibility of AI initiatives. They often force difficult trade-offs against higher-order goals, creating a persistent tension between what is ideal and what is achievable.

4.4.1 Speed

A distinct pattern emerged regarding the necessity of speed in both AI model answers and its explanations, where applicable, delineated by the stage of the AI lifecycle. Technical personnel involved in the development phase generally viewed computational speed as a secondary concern. As P2 explained, workflows that are not yet tied to real-time transactional systems allow for significant latency: *“I don’t think we are worried that it comes with[in] 10 seconds... or maybe even 10 minutes? We would press a button in the evening and come in the morning.”*

However, this tolerance disappears once the system transitions to operational use. Participants acknowledged that *“for a user interface, it’s important to give everything fast,”* recognizing that once the system is deployed for operational use, the speed of responsiveness becomes an important user-facing requirement. For front office professionals like P3 and P14, speed is strictly dictated by key performance indicators (KPIs), or metrics that evaluate the quality of service delivered in a specific banking role. In customer support, this can be the time it takes for an employee to resolve a complaint.

P14 noted that financial crime investigators have hard targets to *“clear the cases in one to three days,”* while P1 highlighted that customer service departments operate under strict timeframes for interaction. Consequently, latency becomes an adoption barrier; P1 warned that if a model *“would take one hour, I don’t think anyone would use it. They would rather type.”* This observation aligns with Kruse, Wunderlich & Bech (2019), who argue that if an AI tool (such as a chatbot) lacks speed, its “added value is questionable.”

P17 described the pressure from internal clients, noting a divide between technical stakeholders who want depth, and operational users who demand efficiency: *“[they say] make it work for me because I don’t have time for bad information.”* Beyond computational speed, Kuiper et al. (2019) also highlight a bureaucratic temporal constraint: organizations worry that using more complex models will trigger lengthy, resource-intensive approval processes with supervisory authorities, which can discourage their adoption.

4.4.2 Cost

Closely linked to the computational demands of speed is the factor of cost. As P1 noted, budgetary constraints are strict and dictate feasibility: *“if we don’t have the budget for that, it will simply not fly.”* Even though other participants were not exactly privy to the financial sheets of the company, most still emphasized the necessity of a rigorous cost-benefit analysis (P5, P7, P13, P16) which changes over

time. This aligns with the concept of “hidden technical debt” in ML systems, the accumulation of unseen maintenance and integration burdens behind apparently simple models (Sculley et al., 2015), suggesting that while rapid deployment offers immediate gratification, it often obscures long-term maintenance challenges and infrastructure costs. As P5 argued, high investment *“must be traded with the benefit you get out of it.”* For example, in the case described by P6, the post-implementation of black-box models for alternative credit scoring resulted in a 40% expansion of the bank’s credit card base, bringing in approximately \$5 million in transactions.

However, the weight given to cost is heavily contingent on the scale and complexity of AI models deployed. P4 qualified that cost becomes critical when dealing with computationally intensive tasks, such as counterfactual explanations which require *“a very expensive search if it’s a big model.”* This distinction was reinforced by participants from the banking sector. P2 noted that their smaller scale of AI application meant they did not incur *“major costs from running explanations,”* whereas P7, operating at a larger volume, emphasized the need to be *“cost efficient because the cost/call centers really look at how many API calls they do.”* Finally, P17 offered a temporal perspective, suggesting that attention to cost fluctuates with the *“hype cycle,”* the pattern where enthusiasm and rapid adoption are later followed by a more sober focus on long-term costs and sustainability. While organizations may ignore expenses during early phases of innovation, P17 observed a shift toward a *“cost-minded”* reality where stakeholders must quantify efficiency and manpower savings: *“You cannot ignore it... otherwise it bites you back.”*

4.5 Synthesizing the Financial AI Trilemma

Across these themes, finance professionals do not experience explainability as a simple trade-off with model accuracy. Instead, our interviews suggest a configuration in which accuracy and compliance are treated as non-negotiable prerequisites, while ease of understanding determines whether systems are usable, defensible, and worth adopting in practice. Cost and speed remain salient, but they are primarily framed as operational constraints and adjustable levers rather than as core criteria in their own right.

We refer to this configuration as the financial AI trilemma that emerged from the interviews. As summarized in Figure 5, accuracy, compliance, and ease of understanding form three interdependent prerequisites for deploying AI in finance, bounded by cost and speed as feasibility constraints. This trilemma captures how practitioners navigate the competing demands of performance, regulation, and intelligibility when deciding whether and how to implement AI systems.

5 Conclusion and Recommendations

Our interviews indicate that explainability in financial AI is not governed by a simple binary trade-off with accuracy, but by a trilemma (Figure 5) in which accuracy and compliance act as non-negotiable constraints and ease of understanding determines whether systems are usable and defensible in practice. We term this configuration the financial XAI trilemma, emphasizing that explainability cannot be optimized in isolation from regulatory and organizational demands drawing from a sociotechnical perspective (Glaser et al., 2021; Kudina & Poel, 2024). Other factors, such as speed and cost, continue to matter but are often deprioritized or treated as adjustable levels whose relative importance shifts depending on the comparison set or operational context. These findings challenge the dominant academic framing that positions explainability merely as a sacrifice of accuracy, overlooking the regulatory and organizational realities that shape model deployment in finance.

This study is exploratory and qualitative in nature. Its aim is not statistical generalization but the description of the experiences of financial professionals. Self-reported experiences cannot be independently verified and may under-represent silent failures (for example, abandoned projects or unreported compli-

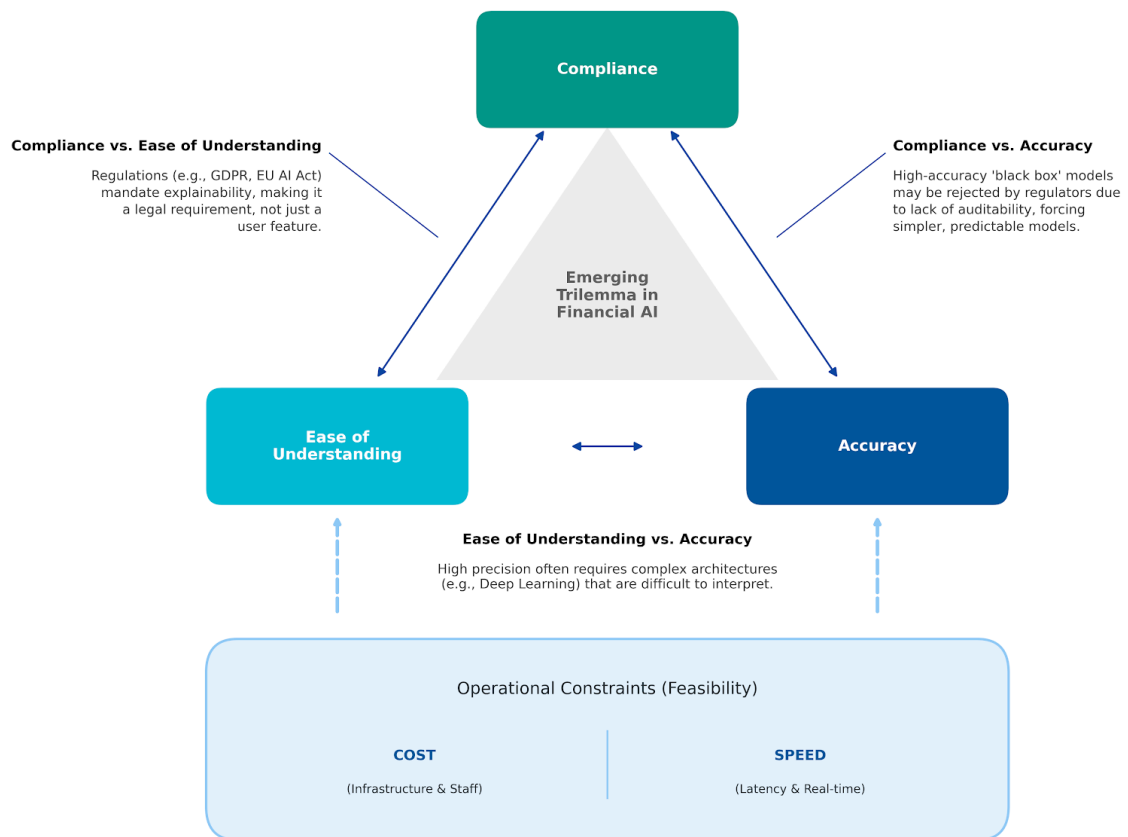


Figure 5: The Trilemma in Financial AI

Source: Authors'; constructed from findings

ance tensions). Moreover, as one participant suggested, one of the criteria discussed may be ranked last only because of the way options are presented to interviewees. The sample is also largely composed of European institutions and reflects the broader gender imbalance in data intensive financial roles. Future work should therefore extend this study to more diverse regions and a more gender-balanced pool of participants in order to examine whether prioritization patterns differ across regulatory environments and professional demographics. These limitations point to the need for systematic, multi-method follow-up studies, especially a deeper investigation into how automated decisions are explained to achieve greater ease of understanding by various audiences.

From a legal and regulatory perspective, framing explainability as a financial XAI trilemma also suggests that explainability requirements should be articulated less as generic transparency mandates and more as context-sensitive obligations that reflect model risk, audience, and use case. This aligns with the recent EU Parliament (2025) resolution on the impact of AI on the financial sector, which highlights the need to balance AI-driven innovation with consumer protection and financial stability, and to avoid overlapping or inconsistent requirements that undermine legal certainty and uptake. Rather than prescribing a single type of explanation, regulators and legislators could require firms to document how they balance accuracy, compliance, and ease of understanding for different stakeholders (e.g., internal risk committees, supervisors, and affected clients), and to justify these choices *ex-ante* in model governance processes. This points toward process-based obligations (e.g., traceable documentation of trade-offs, standardized explanation templates for high-risk models, and early involvement of legal and compliance teams) so that supervisory dialogue focuses not only on whether a model is “explainable” in the abstract, but on whether the explanations offered are adequate and proportionate to the decisions they support.

On a practical note, the study suggests that explanation tools must be designed with simultaneous attention to accuracy, compliance, and ease of understanding, rather than optimizing only one dimension, similar to the trustworthiness framework of Fritz-Morgenthal, Hein and Papenbrock (2022), who combine metrics for explainability and fairness as major objectives, with prediction accuracy and speed as secondary benchmarks. The purpose of this is to evaluate the impact of complex models on human lives and businesses, rather than focusing solely on their design or sophistication. Achieving this requires the involvement of compliance officers and business users early in the model lifecycle and not only during validation, so that explanation requirements are built into systems from the outset and aligned with real operational needs.

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Appendix

A Background of Interview Participants

#	Gender	Type of Institution	Function
P1	F	Bank A	Technical Project Manager
P2	M	Bank A	Technical Project Manager
P3	F	Bank A	Non-technical Manager
P4	M	Bank A	Developer
P5	M	Bank B	Leadership and Strategy
P6	M	Bank C	Developer / Technical Project Manager
P7	M	Bank D	Technical Project Manager
P8	M	Bank D	Technical Project Manager
P9	M	AI Vendor A	Leadership and Strategy
P10	M	AI Vendor B	Non-technical Manager
P11	F	AI Vendor C	Technical Project Manager / Internal Regulator
P12	M	Consultancy A	Analyst
P13	F	Consultancy B	Leadership and Strategy
P14	F	Electronic Money Issuer A	Analyst / Internal Regulator
P15	M	Development Bank A	Analyst
P16	M	AI Vendor D	Leadership and Strategy
P17	M	Bank B	Technical Project Manager
P18	M	Bank E	Leadership and Strategy
P19	M	AI Vendor E	Leadership and Strategy
P20	M	Development Bank B	Analyst

Table 5: Interview participant demographics and background information.

Note: To avoid double counting, each participant was assigned to one functional category only, based on their highest or most overarching role within their organization. When participants held hybrid or overlapping responsibilities (e.g., developer and project manager, or manager and internal regulator), we classified them according to the function that best captured their seniority, decision-making authority, or regulatory relevance. For example, P6 was coded as a Technical Project Manager rather than a developer, and P11 was coded as an Internal Regulator rather than a project manager.